

PROJECT MANAGEMENT (SPECIALIST CLASS 1)

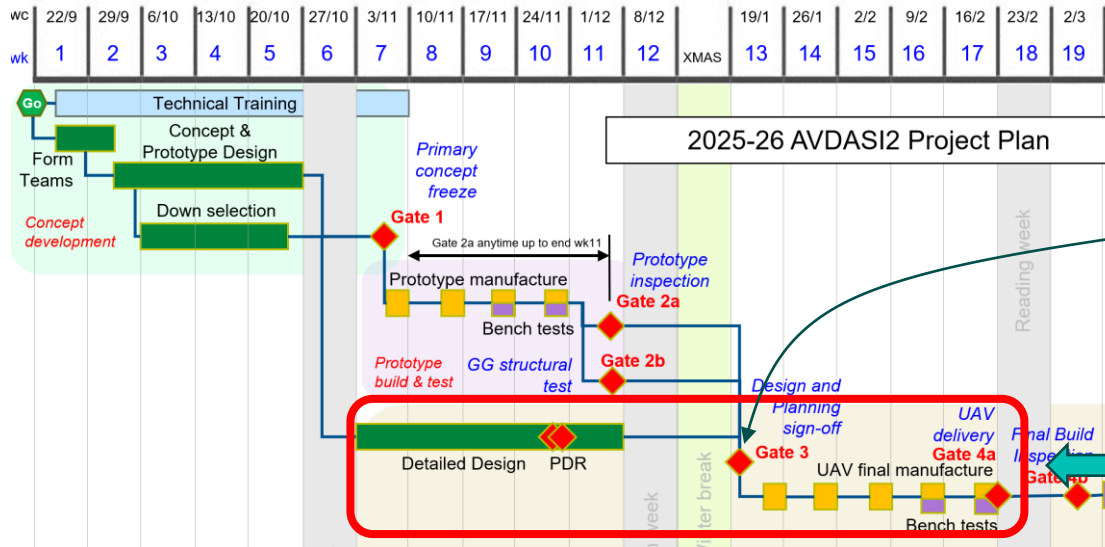
12th November 2025

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Project Management Summative Assessment

- You will submit and discuss your team's project plans for the AVDASI2 project
- These will include detailed/thorough planning for the *Detailed Design and Final manufacture phase*. We expect a: WBS, Network Diagram and Gantt chart
- You will need to show how your detailed plan fits into the larger team and then company plan



Formative feedback is available at this point

Congruent documentation for this phase

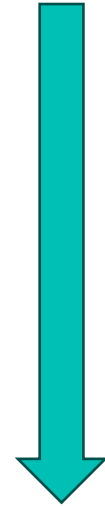
Planning challenges

1. Real-life complexity
2. Limitations (and availability) of software
3. Non-continuous working (Man-days $\neq$ days taken)
4. Types of precedence relationship
5. Resource scheduling (and availability)
6. Intra-team dependencies
7. Representing the macro & micro scale

Increasing planning competence (and difficulty)

WBS + Network + Gantt

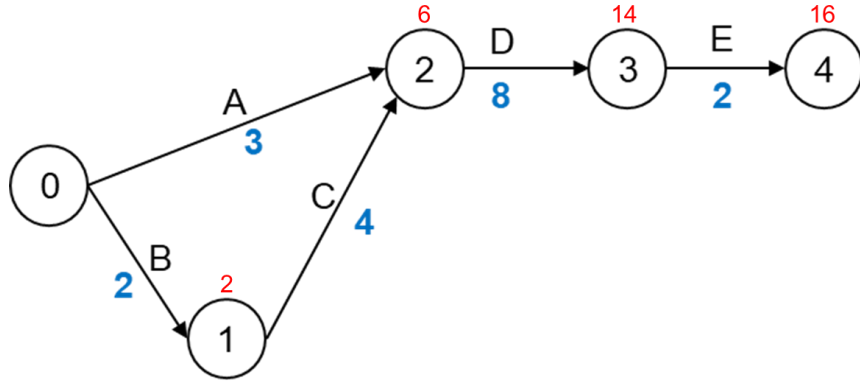
1. The team's final design and manufacturing work is fully defined
2. Tasks are same between the three tools
3. Dependencies are shown on the network
4. Dependencies are shown on the Gantt
5. Intra-team dependencies (i.e. inputs / outputs) are shown on the network and Gantt
6. Network shows precedence relationship type and CP
7. Network shows timings
8. Network accounts for non-continuous working
9. Network and Gantt timings correlate
10. Network and Gantt consider resource scheduling



≥ 7 are exponentially more challenging

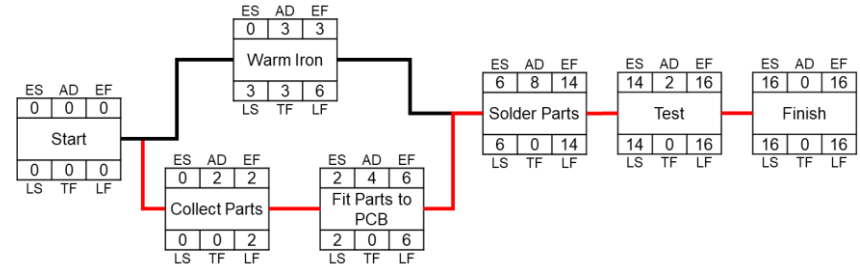
AoA vs AoN (Precedence Diagramming Method PDM)

Activity-on-arrow



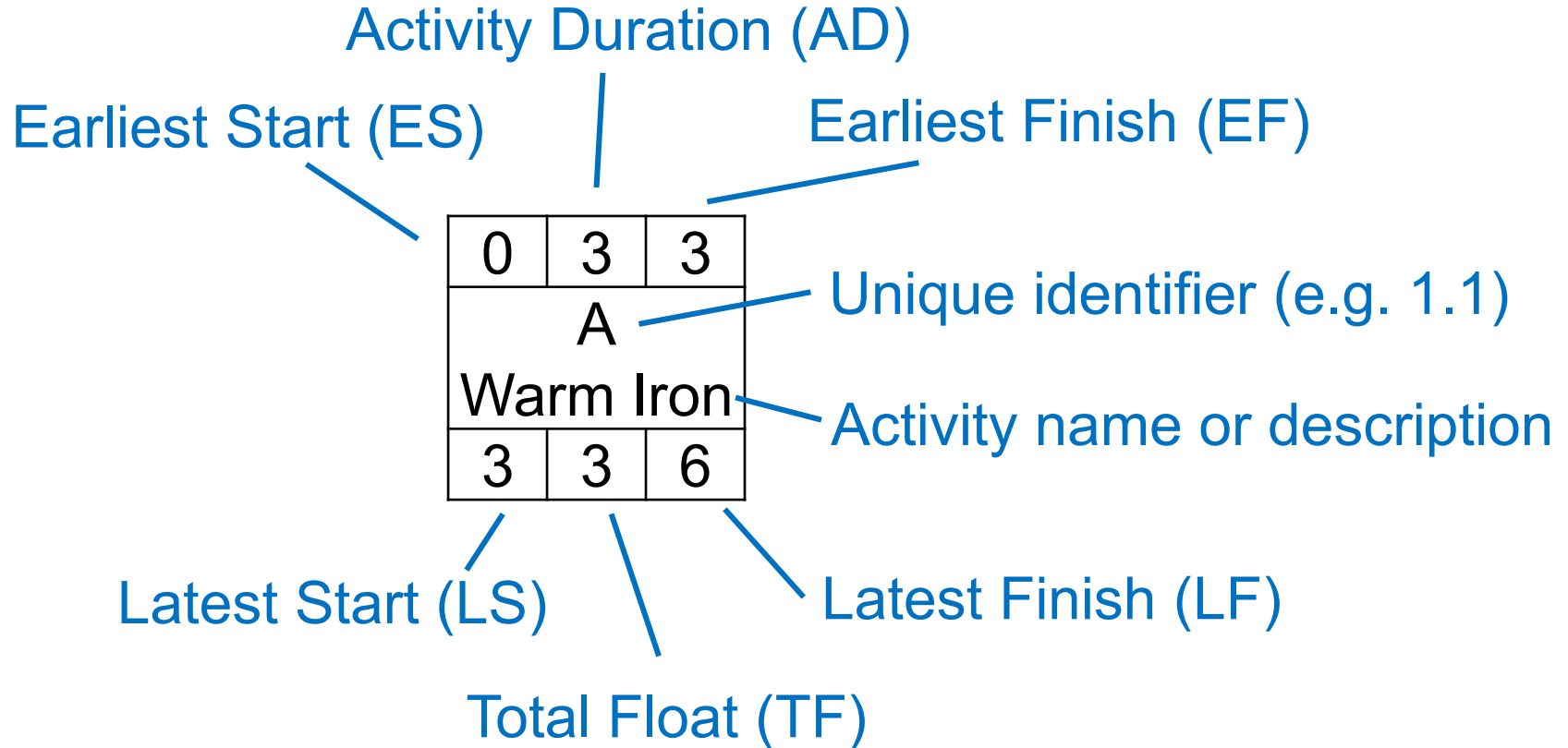
- Arrow shows activity
- Easy to learn
- Difficult to plot for a complex network
- Often requires dummy tasks

Activity-on-Node (PDM)

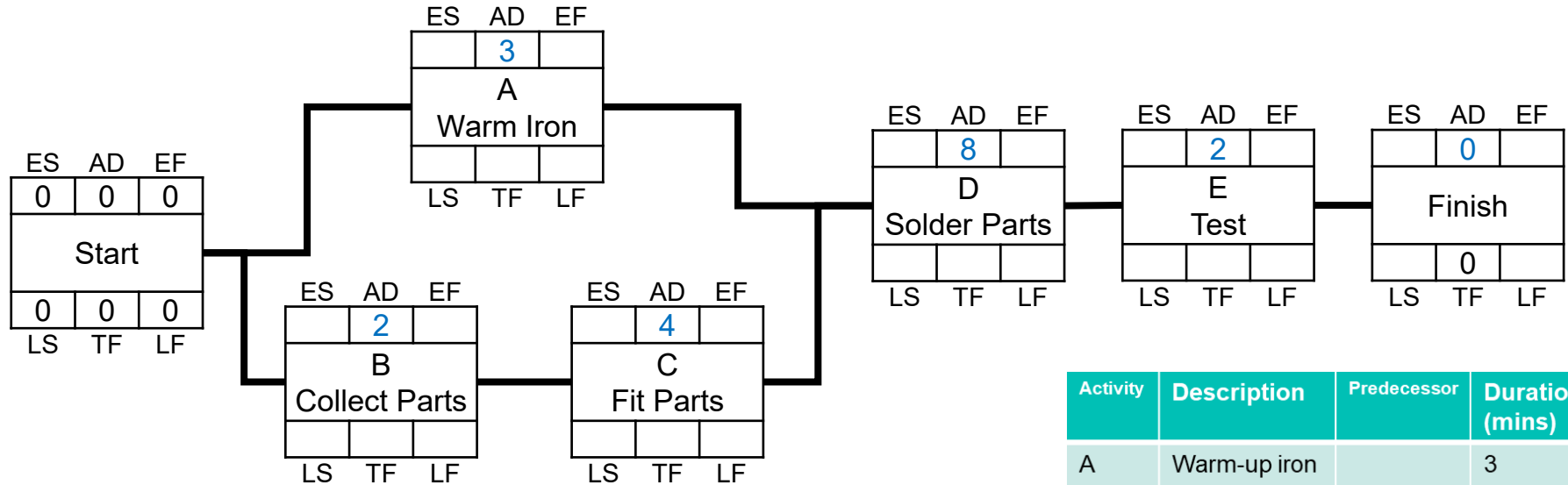


- Node shows activity
- Most popular method
- Shows more timing data
- Easier to plot
- Can encode the maths

Building an Activity-on-node (PDM)



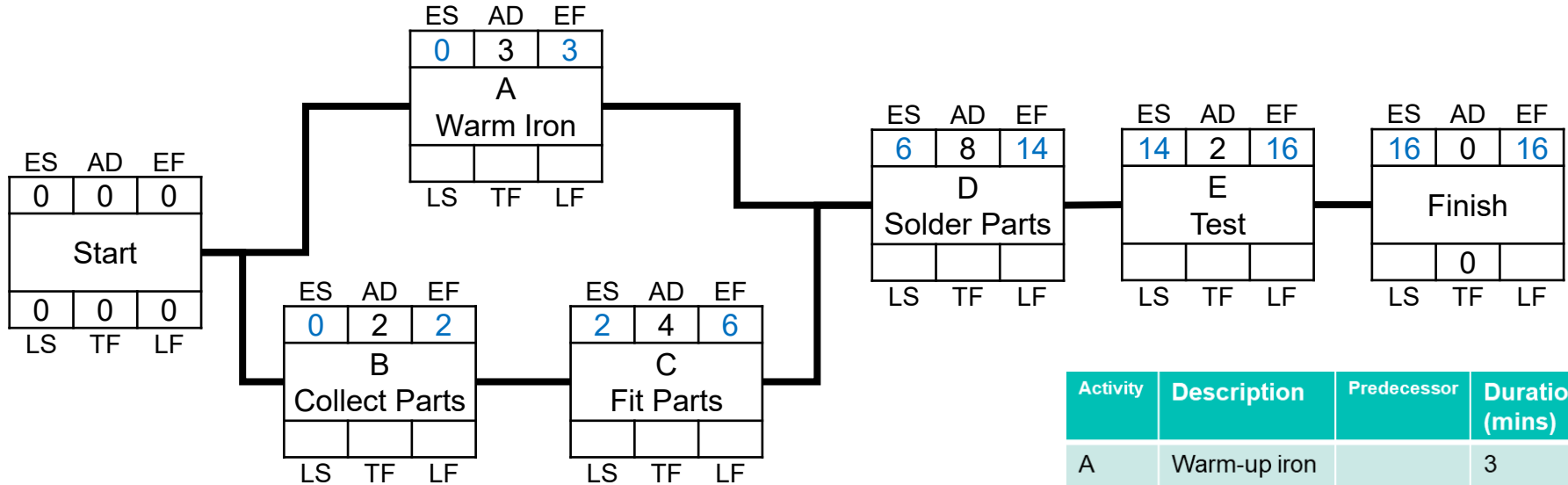
Soldering example



Step 1 – Add durations

Activity	Description	Predecessor	Duration (mins)
A	Warm-up iron		3
B	Collect all parts		2
C	Fit parts to PCB	B	4
D	Solder parts	A,C	8
E	Test	D	2

Soldering example – Forwards Pass

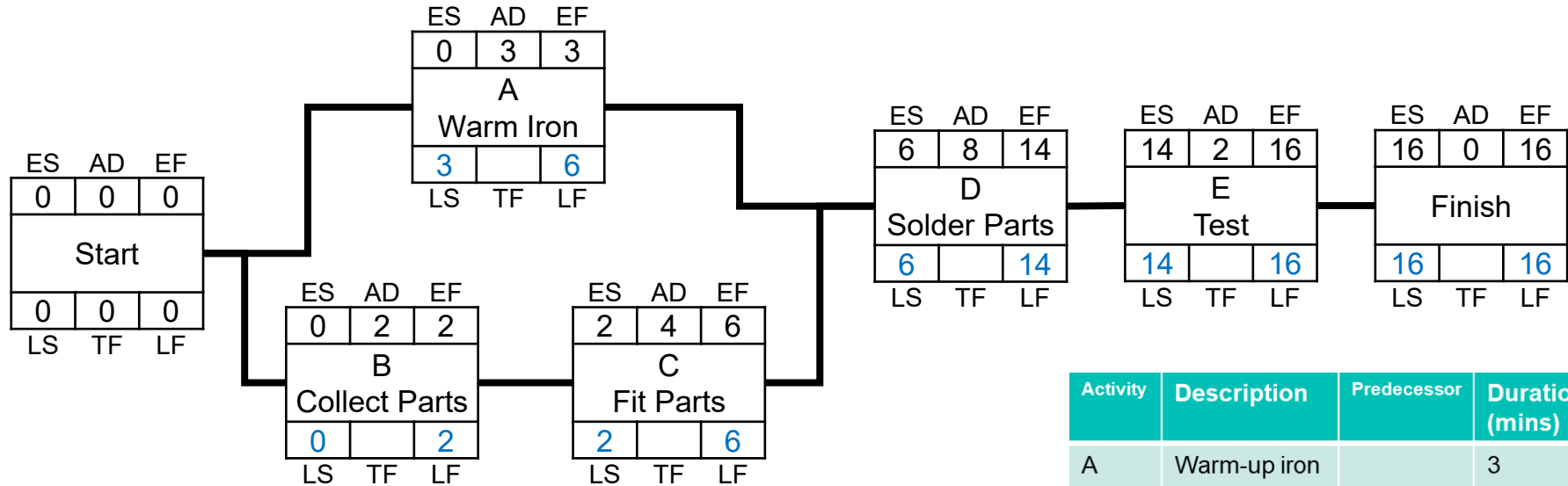


Step 2 – Forward Pass

- $\text{Earliest Start} = \text{Earliest Finish}$ of predecessor activity
(choose latest EF if multiple predecessors)
- $\text{Earliest Finish} = \text{Earliest Start} + \text{Activity Duration}$

Activity	Description	Predecessor	Duration (mins)
A	Warm-up iron		3
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Soldering example – Backwards Pass

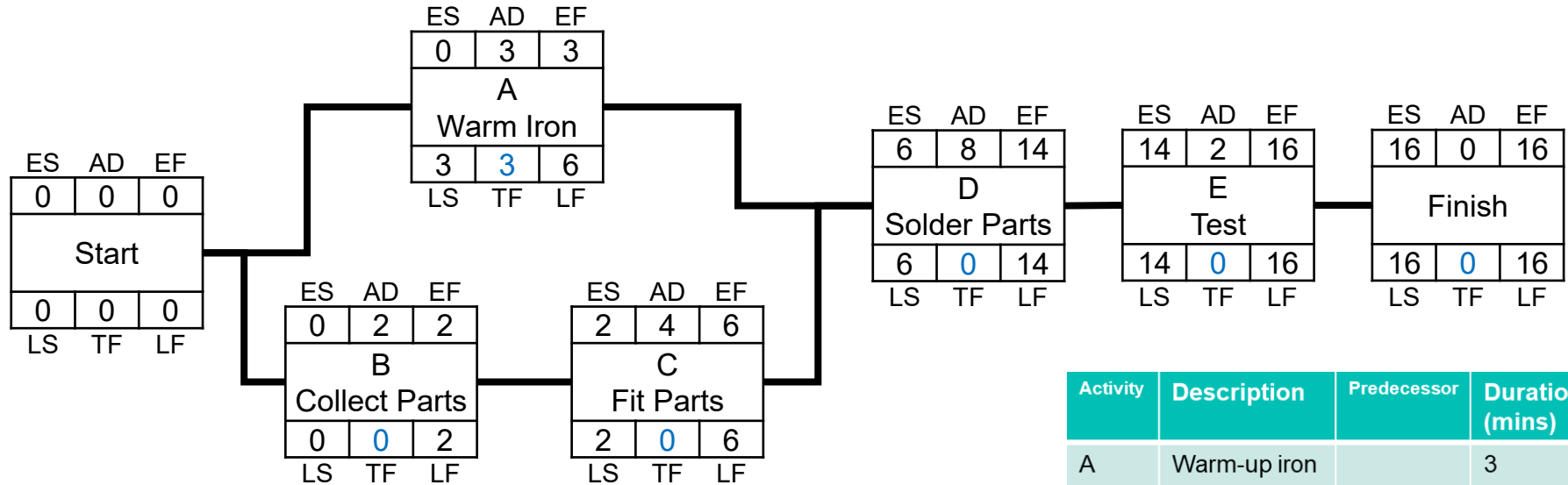


Step 3 – Backwards pass

- Latest Finish* = *Latest start* of successor activity
(choose earliest LS if multiple successors)
- Latest Start* = *Latest Finish* - *Activity Duration*

Activity	Description	Predecessor	Duration (mins)
A	Warm-up iron		3
B	Collect all parts		2
C	Fit parts to PCB	B	4
D	Solder parts	A,C	8
E	Test	D	2

Soldering example – Calculate float

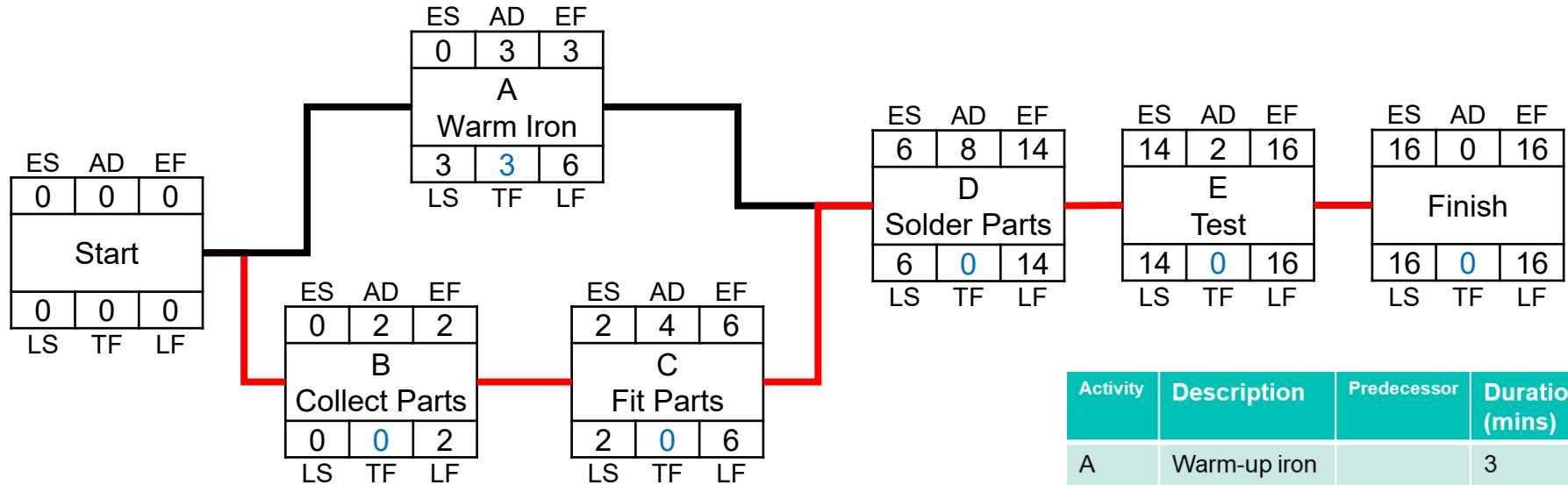


Step 4 – Calculate float

Total Float = *Latest Start* – *Earliest Start* (if 0, then this is a Critical Path task)

Activity	Description	Predecessor	Duration (mins)
A	Warm-up iron		3
B	Collect all parts		2
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Soldering example – Identify CP



Step 4 – Identify Critical Path

Total Float = *Latest Start* – *Earliest Start* (if 0, then this is a Critical Path task)

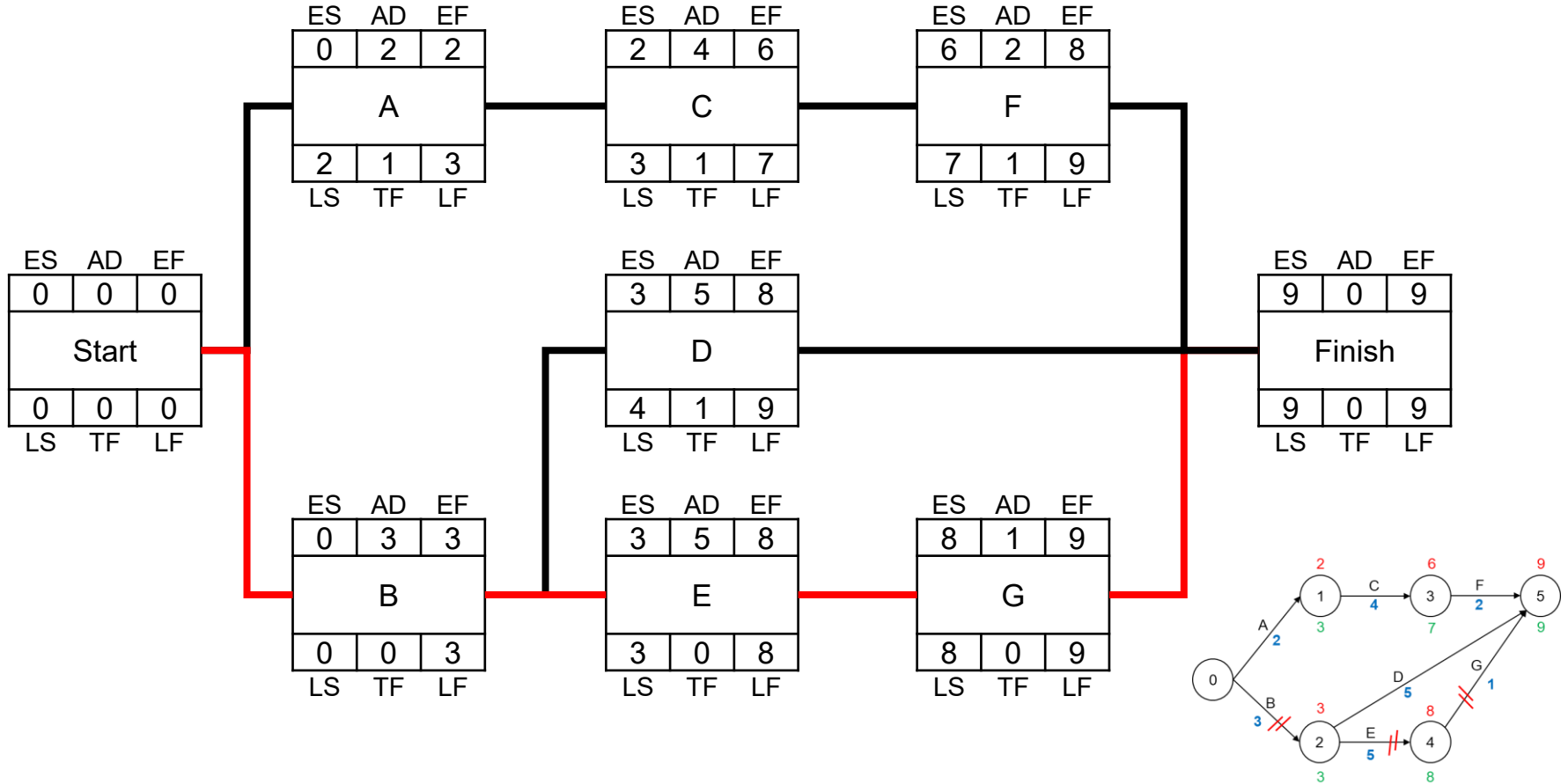
Activity	Description	Predecessor	Duration (mins)
A	Warm-up iron		3
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Building an activity on node (PDM) diagram

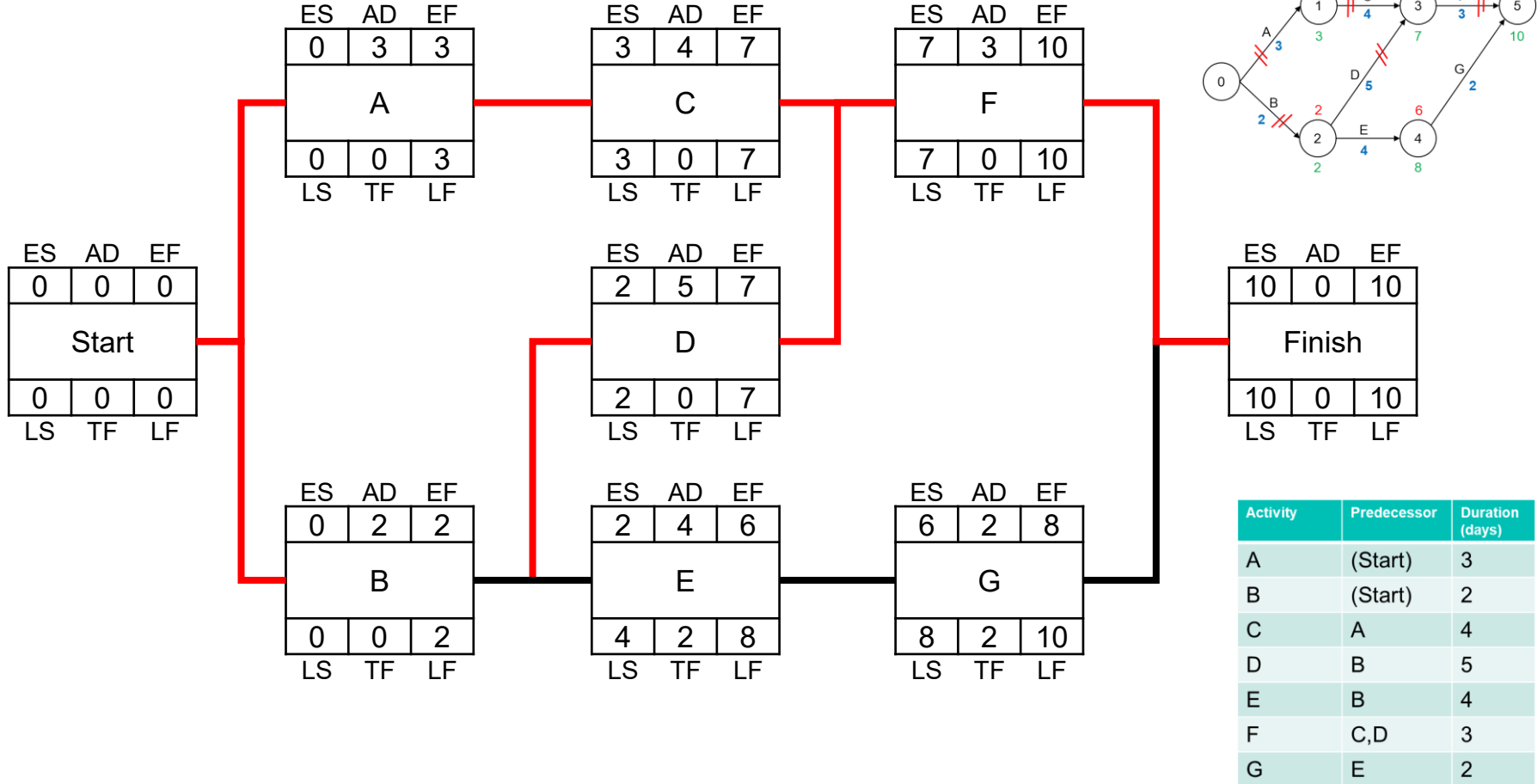
1. Identify the activity dependencies
2. Draw the activity boxes and record a unique identifier and activity description
3. Draw the dependency lines between the boxes
4. Record the *Activity Durations*
5. ►► Forwards Pass: Working through the complete network from start to finish (left to right)
 - a) *Earliest Start* = *Earliest Finish* of predecessor activity (choose latest EF if multiple predecessors)
 - b) *Earliest Finish* = *Earliest Start* + *Activity Duration*
6. ◀◀ Backwards Pass: Working back through the complete network from finish to start (right to left)
 - a) *Latest Finish* = *Latest start* of successor activity (choose earliest LS if multiple successors)
 - b) *Latest Start* = *Latest Finish* - *Activity Duration*
7. *Total Float* = *Latest Start* – *Earliest Start* (if 0, then this is a Critical Path task)
8. Label critical path (bold/colour/symbol)

ES	AD	EF
0	3	3
A Warm Iron		
3	3	6
LS	TF	LF

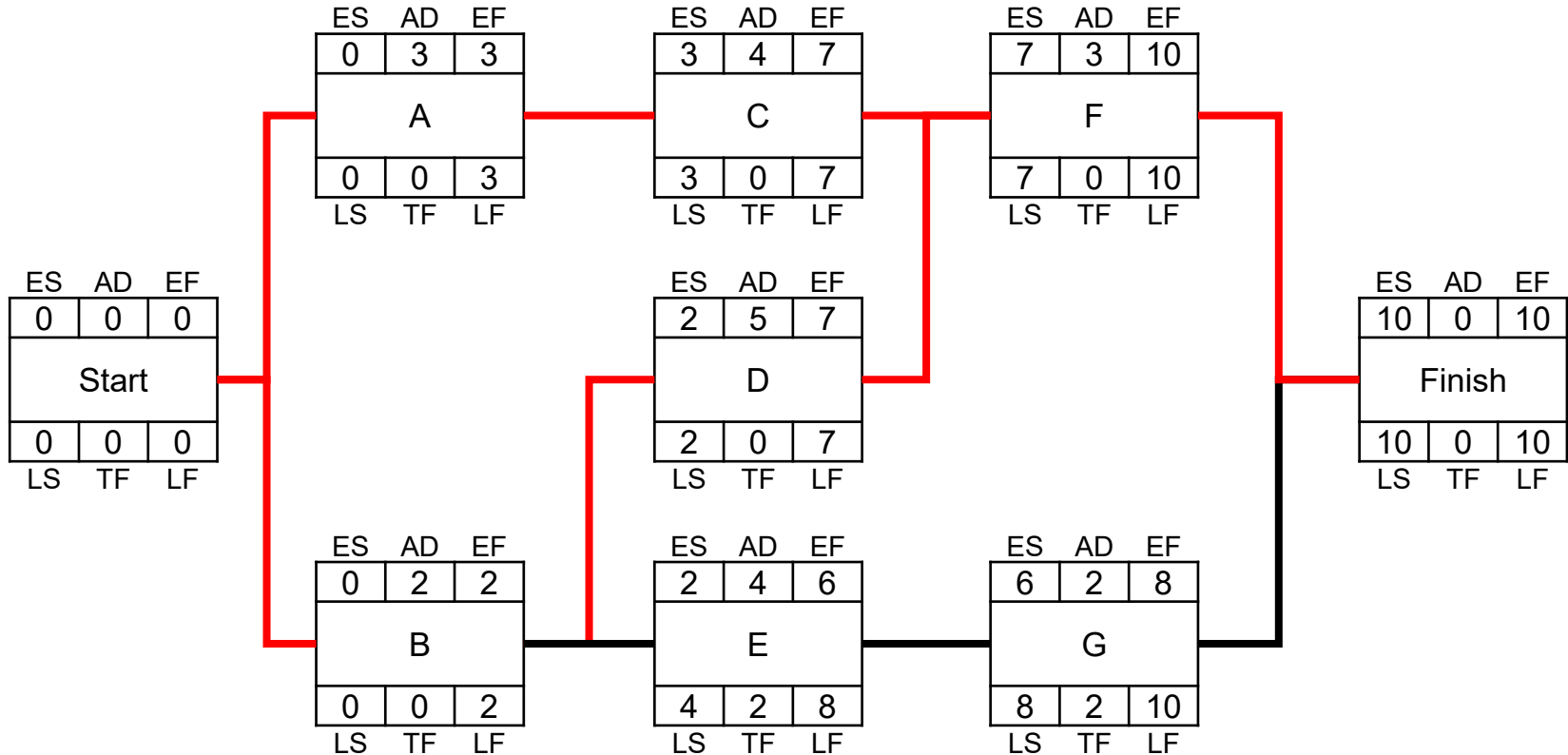
Example 2



Example 3

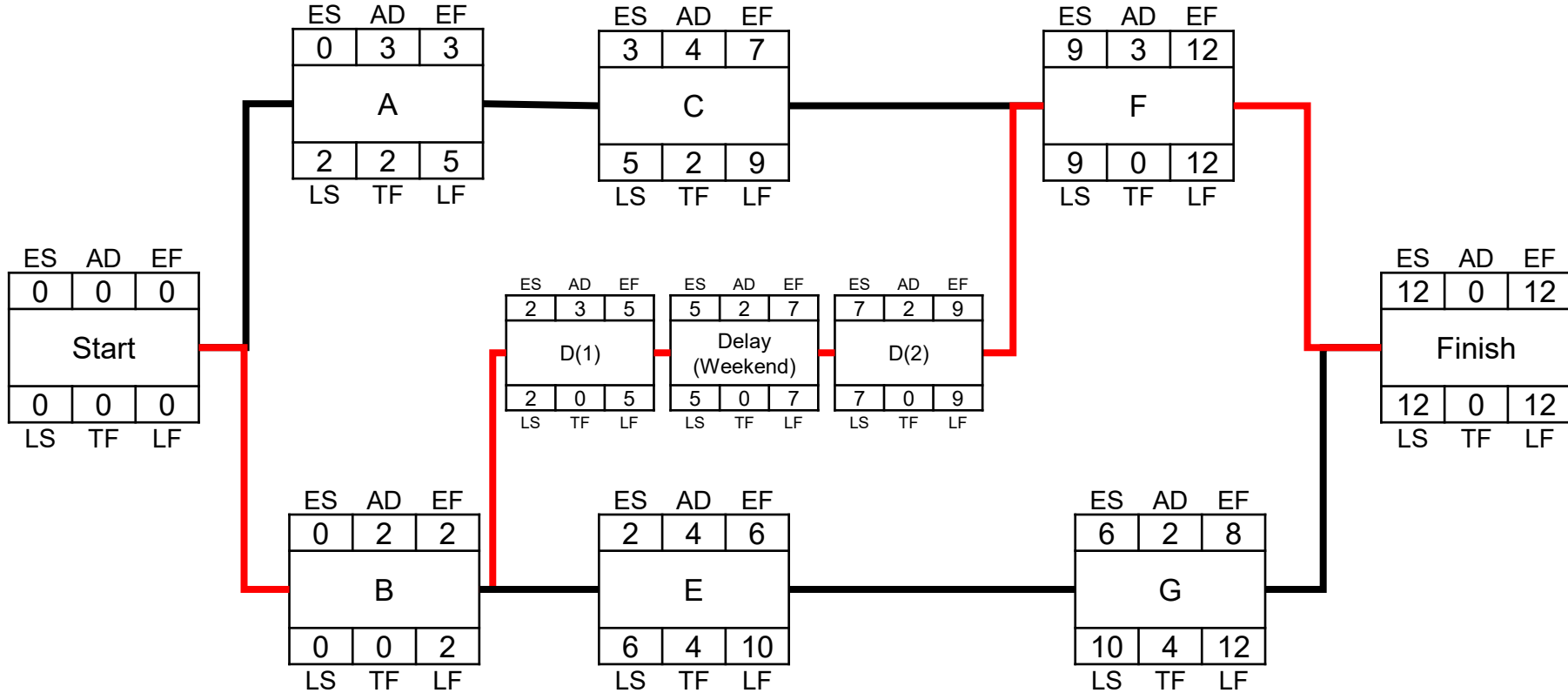


Dealing with non-continuous working



What happens if task D takes 5 days, but bridges a weekend?

Dealing with non-continuous working

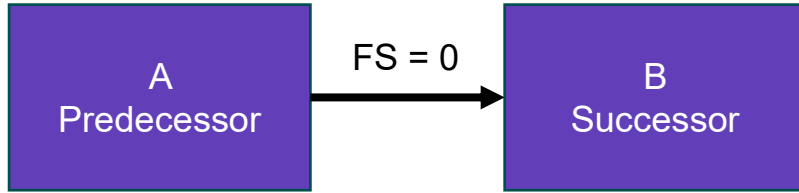


(The whole network might need recalculating multiple times).

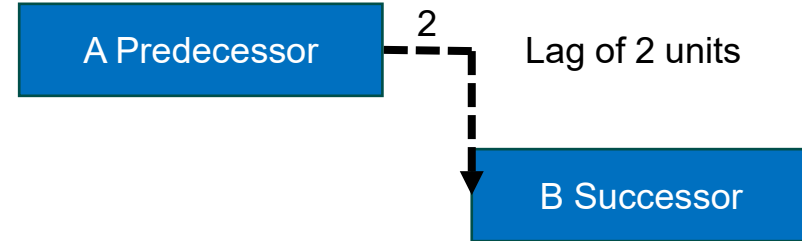
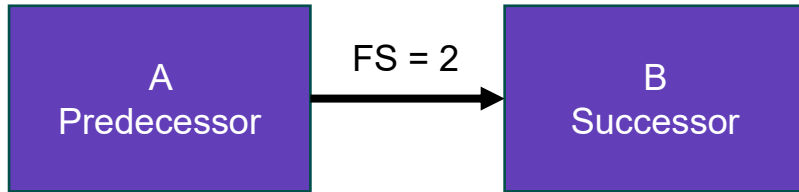
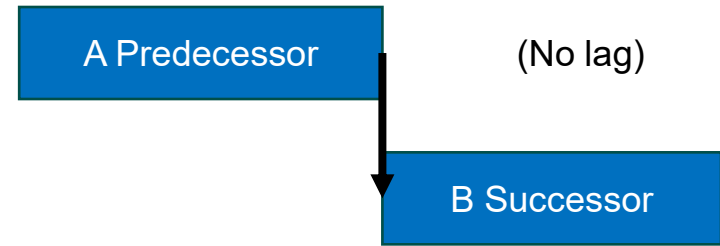
Types of *precedence relationship*

Finish-to-Start (FS)

Network



Gantt



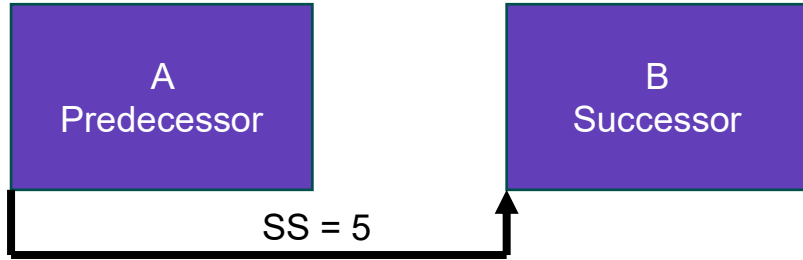
Task B can start 2 days after Task A has finished.

Example – Concrete can be drilled 2 days after it has been laid.

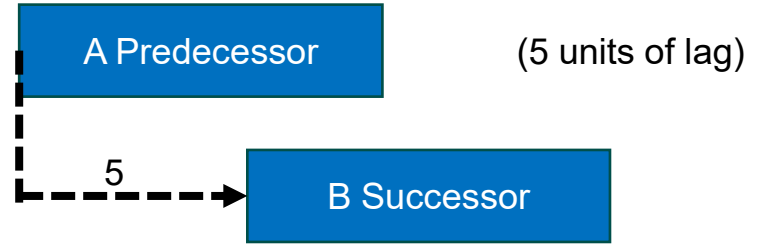
Types of *precedence relationship*

Start-to-Start (SS)

Network



Gantt



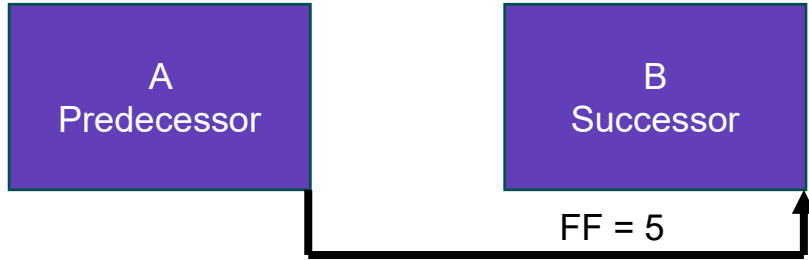
Task B can start 5 days after Task A has started.

Example – Design of actuator mounting bracket can start 5 days after mechanism design has started.

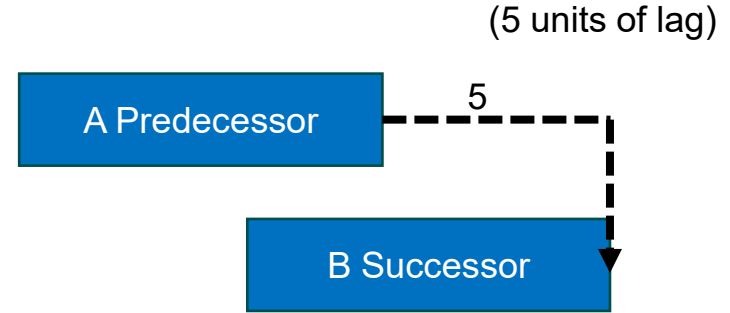
Types of *precedence relationship*

Finish-to-Finish (FF)

Network



Gantt



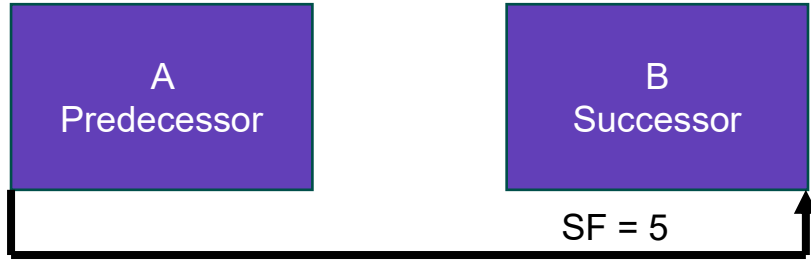
Task B must finish 5 days (at the latest) after Task A has finished.

Example – Must finish painting the parking lines no more than 5 days after finishing asphaltting the road.

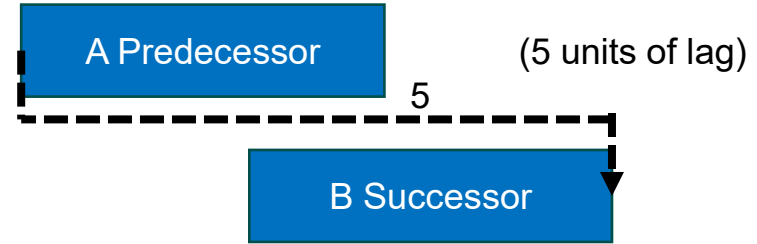
Types of *precedence relationship*

Start-to-Finish (SF)

Network



Gantt



Task B cannot finish until 5 days after Task A has started.

Example – *Phase out old system* cannot finish until 5 days after *testing of new system* has started.

Gantt relationship examples

